

Spectral Tree Inference from Incomplete Matrices: Fiedler Vector Bounds under Sub-sampling

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Abstract

Top-down reconstruction of latent binary phylogenetic trees is fundamentally bottlenecked by the $\Theta(m^2)$ cost of computing pairwise similarities across m leaves. While observing only a sparse, Bernoulli-sampled subset of entries offers a computational workaround, real-world tree imbalance severely degrades the spectral signal, rendering standard balanced-cluster recovery guarantees inadequate. In this work, we frame spectral tree inference as an eigenvector perturbation problem within the matrix-completion regime, showing that the split obtained from the full similarity matrix of an imbalanced latent tree is recovered exactly from the sign pattern of the empirical Fiedler vector computed on a sparse sample. That split is a clan of the tree for any topology.

Our primary contribution is a sharp finite-sample guarantee proving that a sampling rate of $p = \Theta(\log m/m)$ suffices for exact recovery of that split at a fixed imbalance. By utilizing a coordinate-wise Neumann-series expansion rather than a standard Davis-Kahan ℓ_2 transfer, we eliminate a spurious $\sqrt{m}$ factor in the entry-wise error bound. Furthermore, we identify the imbalance ratio η as the governing topological parameter and prove it dictates recovery through two distinct failure mechanisms: artificially inflating subspace coherence (rendering recovery statistically unaffordable) and eroding the spectral gap (rendering recovery structurally impossible). This yields an explicit sample complexity dependence of $(1 + \eta)^3$. Finally, we establish a parallel spectral recovery principle for double-centered distance matrices. Simulations on closed-form block models and Kingman coalescent trees with JC69 sequence evolution validate the predicted $\Theta(\log m/m)$ scaling and the isolated degradation mechanics of topological imbalance.

1 Introduction

A latent tree graphical model relates m observed leaves through unobserved internal nodes, and the inference problem is to recover the topology from leaf data alone [2, 26]. Phylogenetics is the canonical instance and supplies our vocabulary: the leaves are extant organisms, the internal nodes their extinct ancestors, the data aligned sequences. Distance-, likelihood- and spectral-based reconstruction are all well developed [25, 14, 19, 22], but datasets now reach tens of thousands of leaves, and the accurate methods are the slow ones.

That pressure produced the top-down spectral family, which is spectral clustering run on a phylogeny. Build a similarity matrix S over the leaves, form the Laplacian $L = D - S$, split the leaf set by the sign pattern of the Fiedler vector of L , recurse on each side, and merge [2, 30, 16]. The split is provably consistent: the two sides are complementary *clans*, the unrooted analogue of a clade [31], which is what keeps the merge tractable. Both the error and the cost of the recursion land on the top split, since no parent call corrects it and it is the only call that sees all m leaves. Every entry of S is a sequence-pair estimate over a length- ℓ alignment rather than a lookup, so that one call costs $\Theta(m^2)$ estimator evaluations — the cost divide-and-conquer exists to avoid.

This paper asks what happens to the spectrum when that cost is not paid. Compute only a random p -fraction of the pairs, reweight the survivors by $1/p$, and take the Fiedler vector of the incomplete matrix, in the manner of low-rank matrix completion [9, 10, 8, 24, 20, 6]. Does its sign pattern still recover the split, and how small can p be? There is reason to expect an answer well below 1: under a molecular clock the population matrix has exactly two blocks and the Fiedler vector is constant on each clan, so the signal being sampled is extremely concentrated.

Two things keep this from being routine. What decides recoverability from a random subset is how spread the leading subspace is across coordinates, not how large the matrix is [11]; and guarantees for recursive spectral bipartitioning are stated for clusters of comparable size [4, 23], whereas real phylogenies are markedly more imbalanced than standard null models predict [7, 32, 21, 27, 3]. Imbalance pushes both governing quantities the wrong way: it concentrates the split signal onto the smaller clan, raising the sampling cost, and it shrinks the eigenvalue gap that isolates the Fiedler vector, degrading the signal itself. The two failures are distinct — one leaves the eigenvector informative but unaffordable to sample, the other leaves nothing worth sampling — so pooling them shows one curve where there are two mechanisms.

Recovering a split from a sign pattern is a claim about *every* coordinate of an eigenvector, whereas classical perturbation theory controls only its Euclidean norm [12]; passing between the two charges every node as though it were the worst one, which here decides whether a guarantee is satisfiable at all. We therefore work coordinate-wise [13, 15, 1, 29].

Our contributions are:

- We make the imbalance ratio η the single parameter through which tree shape reaches the linear algebra, and express both governing spectral quantities in it: the subspace coherence in closed form, and a lower bound on the Fiedler gap (Section 5).
- We prove that a sampling rate of order $\log m/m$ suffices, at fixed imbalance, for the sign pattern of the empirical Fiedler vector to reproduce the full-data split exactly with high probability, with an explicit cubic dependence on η (Section 4).
- The proof works coordinate-wise rather than in Euclidean norm, through a Neumann expansion and a per-node Bernstein bound (Section 6); on this model the Euclidean route costs a further factor of m in the required rate.
- The same programme on the double-centred *distance* matrix gives a companion statement [18, 5], and we test both in simulation, scoring each sub-sampled split against the one the full matrix returns [28, 17] and stratifying by imbalance rather than pooling.

Section 2 fixes the generative model and states the recovery result that makes the similarity matrix the object of interest; Section 3 gives the cost that motivates sampling, the observation model and the two operators we analyse. Section 4 states the Laplacian rate and ?? the distance rate. Section 5 then routes tree shape into the spectrum through η : it identifies which split the population Fiedler vector returns and bounds the two failure modes. Section 6 assembles the proof and Section 7 tests the rates.

2 Generative model of sequences

This section fixes the generative model in the order in which it is built: the tree, the substitution process on its edges, the clock that turns branch lengths into times, the similarity matrix that the process induces on the leaves, and the block structure that the tree leaves behind in that matrix.

The section closes with the spectral recovery result that makes the similarity matrix the object of interest.

2.1 Tree topology

Let $\mathcal{T} = (V, E)$ be a rooted latent binary phylogenetic tree whose vertex set $V = \mathcal{X} \cup \mathcal{H}$ splits into m observed terminal leaves $\mathcal{X} = \{x_1, \dots, x_m\}$ and $m - 1$ unobserved internal ancestral nodes $\mathcal{H} = \{h_1, \dots, h_{m-1}\}$, among them a distinguished root $r \in \mathcal{H}$ [2]. In the phylogenetic reading the leaves are extant organisms and the internal nodes their extinct ancestors.

Every edge $(h, h') \in E$ carries a *branch length* $\tau_{hh'} \geq 0$, a measure of the distance between the two adjacent nodes it joins. We record branch lengths through absolute node heights: each node h is assigned a height $\tau_h \geq 0$ measured upward from the leaves, so that $\tau_{hh'} = |\tau_h - \tau_{h'}|$ and a node deeper in the past sits higher. For a pair of leaves i, j we write

$$\tau_{ij} := \tau_{\text{MRCA}(i,j)}, \quad d_{ij} := \sum_{e \in \text{path}(i,j)} \tau_e, \quad (1)$$

for the height of their most recent common ancestor and for the total length of the path joining them. Closely related leaves have a low MRCA and a short path; the root is the highest node in the tree. Figure 1a draws the convention.

The two edges descending from r define the *primary split* of the tree: the leaf set $\mathcal{X}$ is partitioned into the two subtrees hanging off the root. This single bipartition $\mathcal{X} = A \cup B$, rather than the full branching pattern of $\mathcal{T}$, is the object we set out to recover; resolving it is the top-down step from which the remaining tree structure follows recursively. We write a and b for the sizes of the two sides, ordered so that $a \leq b$ and $a + b = m$, and we order the leaves along the split, leaves 1 through a on the smaller side, so that each side is a contiguous block of indices.

The two sides of a split have a standard name.

Definition 1 (Clan; adjacent clans). A *clan* of $\mathcal{T}$ is the set of leaves on one side of an edge of the underlying unrooted tree [31, 26]. Two disjoint clans are *adjacent* if their union is again a clan [2], equivalently if the root nodes of the two subtrees share a neighbour.

Figure 1b shows one such cut and the clan it leaves on each side.

The primary split is therefore a pair of complementary adjacent clans, A and B . Rootedness and clans coexist without tension, and it is worth saying why. The model is rooted because the clock and the sampling times introduced below are statements about time, and time has a direction. The recovered object is a clan pair because the similarity matrix of Definition 3 is a function of pairwise distances alone and is therefore invariant to where the root is placed on the tree: the bipartitions it can express are exactly the edge cuts of the unrooted tree. A *clade* is the special case in which the cut edge is incident to the root [31], which is the case the primary split realizes.

2.2 The substitution process

Sequence data arrive by running a Markov substitution process down the edges of $\mathcal{T}$. The process is fixed by a single rate matrix.

Assumption A1 (GTR substitution model). Sequences of length ℓ over a d -character alphabet evolve along $\mathcal{T}$ under a *General Time-Reversible* (GTR) rate matrix $Q \in \mathbb{R}^{d \times d}$ satisfying (i) $Q_{ij} \geq 0$ for $i \neq j$, (ii) $\sum_j Q_{ij} = 0$ for all i , (iii) a stationary distribution $\pi > 0$ with $\pi Q = 0$, and (iv) $\text{diag}(\pi)Q$ symmetric (reversibility).

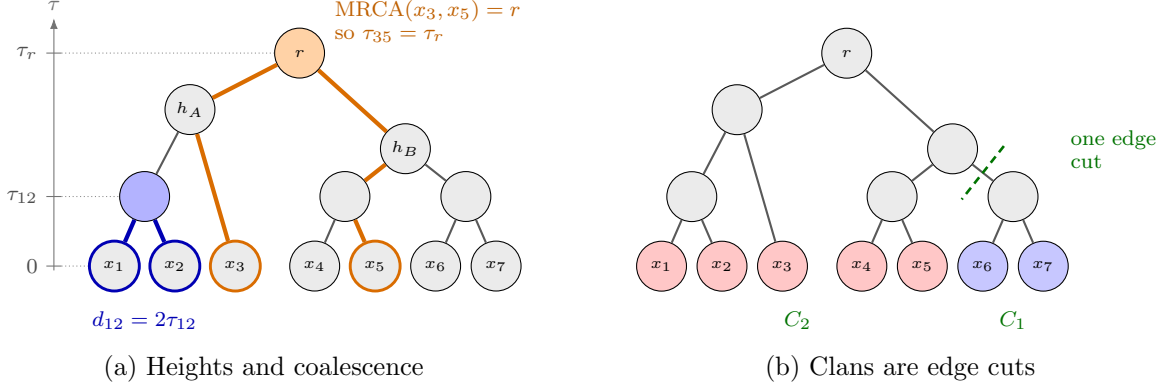


Figure 1: A tree with $m = 7$ leaves. **(a)** The blue path joins a pair that coalesces early: its two legs are equal, so $d_{12} = 2\tau_{12}$, the ultrametric identity (3). The orange path joins a pair whose most recent common ancestor is the root, so $\tau_{35} = \tau_r$; every pair separated by the root behaves the same way, which is what collapses a whole block of S to a single value in Proposition 1. **(b)** Single edge cuts leaves a clan on each side (Definition 1); the cut shown gives $C_1 = \{x_6, x_7\}$ and $C_2 = \{x_1, \dots, x_5\}$.

Conditions (i)–(ii) make Q a generator and (iii)–(iv) make the process reversible; together they imply that all eigenvalues of Q are non-positive, and hence that $\text{tr}(Q) < 0$. For a pair of *adjacent* nodes h, h' the substitution matrix over the branch joining them is the matrix exponential of the generator run for the branch length,

$$P(h \mid h') = \exp(Q \tau_{hh'}), \quad P(h \mid h')_{ba} = \mathbb{P}[h = b \mid h' = a]. \quad (2)$$

By the Markov property the transition matrix is multiplicative along a path: for nodes joined through an intermediate node, $P(h_1 \mid h_3) = P(h_1 \mid h_2)P(h_2 \mid h_3)$. Two nodes separated by total branch length t therefore satisfy $P = \exp(Qt)$ whether or not they are adjacent.

2.3 The molecular clock

So far branch lengths are abstract edge weights. The clock identifies them with elapsed time.

Assumption A2 (Molecular clock with contemporaneous leaves). Branch lengths measure evolutionary time: the substitution process along an edge is determined by Q and by the time difference between its endpoints alone, with no per-lineage rate variation. All m leaves are sampled at the same time, at height $\tau_x = 0$.

Assumption A2 has one consequence, and every structural property below is drawn from it rather than assumed separately. Since all leaves sit at height 0, the path from a leaf i up to any ancestor h has length τ_h ; so for every pair of leaves both legs of the path through their MRCA have the same length,

$$d(i, \text{MRCA}(i, j)) = d(j, \text{MRCA}(i, j)) = \tau_{ij}, \quad \text{hence} \quad d_{ij} = 2\tau_{ij}. \quad (3)$$

A tree with this property is *ultrametric*. Pairwise path length and MRCA height carry the same information, and either may be used interchangeably; we use the height, because it is the quantity the tree topology fixes.

2.4 The similarity matrix

The substitution process is not observed; what is available is a similarity between each pair of leaves, computed from their aligned sequences.

Definition 2 (Pairwise similarity). The similarity between two nodes is the geometric mean of the determinants of the two transition matrices between them [2],

$$S(h_i, h_j) := \sqrt{\det P(h_i | h_j) \det P(h_j | h_i)}, \quad S(h_i, h_i) = 1. \quad (4)$$

Definition 3 (Similarity matrix and similarity graph). The *similarity matrix* $S \in \mathbb{R}^{m \times m}$ collects the similarities over the observed leaves, $S_{ij} := S(x_i, x_j)$, and is symmetric with unit diagonal. It is the weighted adjacency matrix of the *similarity graph* G : the complete graph whose vertices are the leaves of $\mathcal{T}$ and whose edge (x_i, x_j) carries weight S_{ij} [2].

Under Assumption A1 the similarity has a closed form in the tree. For a single branch of length τ , reversibility makes the two transition matrices similar under $\text{diag}(\pi)$, so they share a determinant, and $\det \exp(Q\tau) = \exp(\text{tr}(Q)\tau)$ gives $S = \exp(\text{tr}(Q)\tau)$. Because $\det$ is multiplicative and P is multiplicative along paths, so is the similarity: for any two leaves $S_{ij} = \exp(\text{tr}(Q)d_{ij})$. Substituting the ultrametric identity (3),

$$S_{ij} = \exp(2 \text{tr}(Q) \tau_{ij}). \quad (5)$$

Since $\text{tr}(Q) < 0$, the similarity in (5) decays exponentially in the height of the MRCA, equivalently in the evolutionary time separating the pair: leaves whose common ancestor is recent are similar, and similarity falls off geometrically as that ancestor recedes into the past. The matrix S therefore records the tree's coalescence heights, one entry per pair.

2.5 The topology inside the similarity matrix

The tree is recoverable from S because clans, and only clans, leave a rank signature in it.

Lemma 1 (Clans are the rank-one blocks, [19, Lem. 3.1]). *Let A and B partition the leaves of an unrooted binary tree $\mathcal{T}$, and let $S(A, B)$ denote the $|A| \times |B|$ submatrix of cross-pair similarities. Then $S(A, B)$ is rank-one if and only if A and B are clans of $\mathcal{T}$.*

Lemma 1 is due to Jaffe et al. and is the structural result STDR is built on [19, 2]. It is the precise sense in which the topology of $\mathcal{T}$ is embedded in the structure of S : a candidate bipartition is a split of the tree exactly when the block of similarities it cuts collapses to rank one. Ordering the leaves along the primary split as above, the within-clan pairs occupy the two diagonal blocks of S and the cross-clan pairs its off-diagonal block, and Lemma 1 says that this off-diagonal block is rank-one at the primary split and at no bipartition that is not a split.

Under the clock the off-diagonal block is more than rank-one — it is constant. This is the model the paper works in.

Definition 4 (Clan-Floor Block Model (CFBM)). A symmetric matrix S with unit diagonal satisfies a localized *Clan-Floor Block Model* at a bipartition $\mathcal{X} = A \cup B$ if the cross-clan similarities are constant at a baseline $\beta_0 \in (0, 1)$ while the intra-clan similarities clear that baseline from above by a floor,

$$S_{ij} = \beta_0 \quad (i \in A, j \in B), \quad \beta_0 < S_{\text{in}}^{\min} \leq S_{ij} \leq S_{\text{max}} < 1 \quad (i \neq j \text{ in the same clan}), \quad (6)$$

where $S_{\text{in}}^{\min}$ is the weakest intra-clan similarity and $S_{\text{max}} := \max_{i \neq j} S_{ij}$. The diagonal blocks are otherwise arbitrary. We call the special case in which the two diagonal blocks are themselves constant the *constant-block* CFBM.

Definition 4 is the phylogenetic instance of the constant block model of Balakrishnan et al. [4], in the localized form used for latent trees by [2]: two clusters with a constant between-cluster level, and within-cluster levels constrained only from below.

Nothing further need be assumed: Assumptions A1 and A2 already put S in this model at the primary split.

Proposition 1 (The population similarity is a CFBM at the primary split). *Let S be the population similarity matrix of Definition 3 under Assumptions A1 and A2, and suppose every branch of $\mathcal{T}$ has positive length. Then S satisfies Definition 4 at the primary split $\mathcal{X} = A \cup B$, with*

$$\beta_0 = \exp(2 \operatorname{tr}(Q) \tau_r), \quad S_{\text{in}}^{\min} = \exp(2 \operatorname{tr}(Q) \tau^{\max}) > \beta_0, \quad (7)$$

where τ_r is the root height and $\tau^{\max} < \tau_r$ is the greatest within-clan MRCA height. In particular $S_{ii} = 1$ and $0 < \beta_0 \leq S_{ij} \leq S_{\text{max}} < 1$ for every $i \neq j$, and the block degrees of $L = D - S$ are $D_A = \operatorname{diag}(S_A \mathbf{1}_a) + b\beta_0 I_a$ and $D_B = \operatorname{diag}(S_B \mathbf{1}_b) + a\beta_0 I_b$.

Proof. Let $i \in A$ and $j \in B$. Every path from A to B traverses the root, so $\operatorname{MRCA}(i, j) = r$ and $\tau_{ij} = \tau_r$ for every cross-clan pair; this is purely topological. By (5) all cross-clan entries then equal the single value $\beta_0 = \exp(2 \operatorname{tr}(Q) \tau_r)$, which lies in $(0, 1)$ because $\operatorname{tr}(Q) < 0$ and $\tau_r > 0$.

Now let $i \neq j$ lie in the same clan, say A , with clan root h_A . Then $\operatorname{MRCA}(i, j)$ is a descendant of h_A , so $\tau_{ij} \leq \tau^{\max} \leq \max(\tau_{h_A}, \tau_{h_B}) < \tau_r$, the last inequality strict because the two root branches have positive length. As $\operatorname{tr}(Q) < 0$, (5) is strictly decreasing in τ_{ij} , so $S_{ij} \geq \exp(2 \operatorname{tr}(Q) \tau^{\max}) = S_{\text{in}}^{\min} > \beta_0$. In the other direction, $\operatorname{MRCA}(i, j)$ is a strict ancestor of both i and j for $i \neq j$, so $\tau_{ij} > 0$ and $S_{ij} < 1$; taking the maximum over the finitely many pairs gives $S_{\text{max}} < 1$. Together with $\tau_{ii} = 0$, giving $S_{ii} = \exp(0) = 1$, this is (6) and the stated range. The block degrees follow by summing the rows of S within each block and adding the constant cross-block contribution, $b\beta_0$ for a row of A and $a\beta_0$ for a row of B . $\square$

With the baseline and the floor in hand, the primary split is visible in S as two strong diagonal blocks separated by a weaker constant off-diagonal block, and in $L = D - S$ as the corresponding block form,

$$S = \begin{pmatrix} S_A & \beta_0 J_{a \times b} \\ \beta_0 J_{b \times a} & S_B \end{pmatrix}, \quad L = \begin{pmatrix} D_A - S_A & -\beta_0 J_{a \times b} \\ -\beta_0 J_{b \times a} & D_B - S_B \end{pmatrix}, \quad (8)$$

with $J_{a \times b}$ the all-ones block and β_0 the cross-clan baseline. The diagonal blocks are whatever the two subtrees make them; only the off-diagonal block is pinned.

2.6 Spectral quantities

The split is read out of S spectrally, through the Laplacian.

Definition 5 (Laplacian and Fiedler vector). The *combinatorial Laplacian* of the similarity graph is $L := D - S$, where D is diagonal with $D_{ii} = \sum_j S_{ij}$ [2]. Write $0 = \lambda_1 \leq \lambda_2 \leq \lambda_3 \leq \dots \leq \lambda_m$ for its eigenvalues. The *Fiedler vector* $v^{(2)}$ is a unit eigenvector belonging to the second-smallest eigenvalue λ_2 [16, 30].

Two scalar parameters enter the sampling rate of Section 4. The first is purely topological.

Definition 6 (Imbalance ratio). For clan sizes $a \leq b$ with $a + b = m$, the *imbalance ratio* measures how unevenly the primary split divides the leaves,

$$\eta := \frac{b}{a} \geq 1. \quad (9)$$

A perfectly balanced split has $\eta = 1$, while caterpillar-like lineages reach $\eta = \mathcal{O}(m)$; this parameter plays the role of the balanced-cluster constant of Balakrishnan et al. [4]. The second parameter measures how sharply intra-clan similarities exceed cross-clan ones.

Definition 7 (Structural margin). Write $S_{\text{out}}^{\max}$ and $S_{\text{out}}^{\min}$ for the largest and smallest similarity across the primary split, and $S_{\text{in}}^{\min}$, S_{max} as in (6), so that S_{max} is the largest off-diagonal entry of S overall. Following the range-restriction framework of [4], the *structural margin* is

$$\rho := (S_{\text{in}}^{\min} - S_{\text{out}}^{\max}) - (S_{\text{out}}^{\max} - S_{\text{out}}^{\min}). \quad (10)$$

The first bracket is the separation between the weakest within-clan pair and the strongest cross-clan pair; the second penalizes variation inside the cross-clan block. Under Assumption A2 that variation vanishes, $S_{\text{out}}^{\max} = S_{\text{out}}^{\min} = \beta_0$ by Proposition 1, the penalty collapses, and

$$\rho = S_{\text{in}}^{\min} - \beta_0 > 0. \quad (11)$$

2.7 Spectral recovery of the split

We can now say why the similarity matrix is the object to compute. Thresholding the Fiedler vector of the similarity graph never returns a bipartition that is not a split of the tree.

Theorem 1 (Consistency of the Fiedler split, [2, Thm. 4.2]). *Let G be the similarity graph of a binary tree $\mathcal{T}$ and let v be the Fiedler vector of G . Partition the leaves by the sign pattern of v ,*

$$C_1 = \{i : v_i \geq 0\}, \quad C_2 = \{j : v_j < 0\}. \quad (12)$$

Then C_1 and C_2 are adjacent clans of $\mathcal{T}$.

Theorem 1 carries no hypothesis on tree shape or on the substitution parameters: given the exact similarity matrix, the sign rule (12) always cuts the tree along a single edge. This is what makes top-down spectral recovery viable at all, and it is why the recursion can merge its two subproblems — each side is a genuine subtree. Figure 2 shows the split on three simulated trees of increasing imbalance.

We write $\Pi(S)$ for the bipartition (12) produced by the Fiedler vector of $L = D - S$, so Theorem 1 reads: $\Pi(S)$ is a clan pair of $\mathcal{T}$. This is the object the rest of the paper is about — Section 3 asks at what sampling rate $\Pi(\hat{S})$ still equals $\Pi(S)$ — and it is well-defined whenever λ_2 is simple.

What the theorem does not say is *which* edge of $\mathcal{T}$ it cuts, and it leaves λ_2 free to be degenerate, in which case $\Pi(S)$ is not even well defined. One quantitative condition on the tree closes both points, and it is the only condition beyond Assumptions A1 and A2 that the main result carries.

Condition C1 (Separation margin / recoverability). The structural margin strictly dominates the imbalance-weighted cross-clan boundary,

$$\rho > \eta S_{\text{out}}^{\max}, \quad \text{equivalently} \quad S_{\text{in}}^{\min} > (1 + \eta) S_{\text{out}}^{\max}. \quad (13)$$

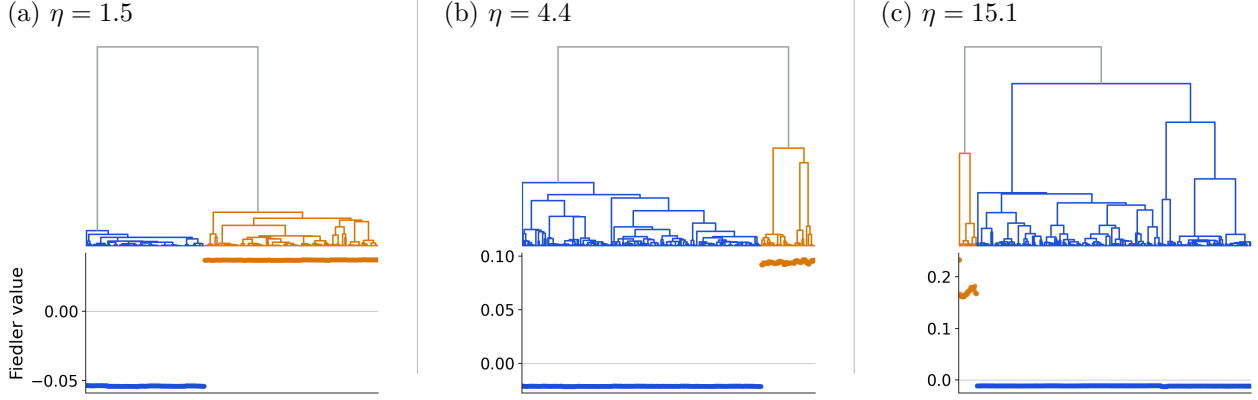


Figure 2: $\Pi(S)$ on three trees, one per panel. Each is drawn from a pool of simulated Kingman trees ($m = 500$ leaves, JC69 sequences of length $\ell = 10^4$) binned by the imbalance its Fiedler split realises. In each panel the tree is on top, its branches coloured by the sign of the Fiedler vector — grey on the one branch that spans both signs — and below it the Fiedler value of each leaf is plotted at that leaf’s own horizontal position in the tree. The sign pattern cuts a single edge every time, as Theorem 1 requires, and the smaller the clan it cuts off, the larger the values carried by its leaves.

Under Assumption A2 that range collapses to the single value β_0 by Proposition 1, leaving $S_{\text{in}}^{\min} > (1 + \eta)\beta_0$.

With the model, the two governing parameters and Condition C1 in hand, Section 3 fixes what is observed and Section 4 states the rate. Section 5 then returns to the population matrix, identifies which clan pair $\Pi(S)$ is, and traces how the imbalance η moves the two spectral quantities that the constants of that rate conceal.

3 Problem setting

3.1 Why a sampled similarity

Theorem 1 makes divide-and-conquer over the similarity matrix a viable strategy for large trees: partition the leaves by the sign of the Fiedler vector, solve each side independently by any reconstruction method, and merge the two subtrees. The recursion is attractive because the expensive part of tree reconstruction is superlinear in the number of leaves, so cutting the problem in half repeatedly reduces the total work sharply, while Theorem 1 guarantees that each cut is a genuine split and the merge is therefore well-posed [2].

The top-level call is the one that governs both the error and the cost. It is the only call that sees all m leaves, and no parent call corrects it: a wrong split at the root cannot be repaired further down. It is also where the similarity matrix is largest.

That is the bottleneck. An entry of S is not a table lookup but a statistic of a pair of sequences: the log-det similarity of Definition 2 is estimated from the joint character counts of leaves i and j over a length- ℓ alignment. Forming the full matrix at the top level therefore costs

$$\binom{m}{2} = \Theta(m^2) \text{ pairwise estimates,} \quad \Theta(m^2\ell) \text{ character comparisons,} \quad (14)$$

which for $m \sim 10^5$ leaves is on the order of 10^{10} pairwise estimates before the recursion has taken

a single step. The quadratic term is exactly the cost that divide-and-conquer was introduced to avoid, and it is paid in full at the root. This raises the question the paper answers: can the top-down split be computed at almost linear cost, with a guarantee that it is still the same split the full matrix would have returned?

3.2 The observation model

Our answer is to compute only a random subset of the pairs. Following the matrix-completion paradigm [10, 9], we observe entries through a random mask and reweight: let $\Omega \in \{0, 1\}^{m \times m}$ be a symmetric mask whose upper-triangular entries are independent Bernoulli(p), $p \in (0, 1]$, and form the empirical similarity by inverse-probability weighting [8],

$$\hat{S}_{ij} = \begin{cases} \frac{1}{p} \Omega_{ij} S_{ij}, & i \neq j, \\ 1, & i = j. \end{cases} \quad (15)$$

Only the $\Theta(pm^2)$ pairs in the mask are ever estimated, so p is the fraction of the cost in (14) that is actually paid, and a rate $p = \Theta(\log m/m)$ would bring the top-level call down to $\Theta(m \log m)$ pairwise estimates.

This estimator is unbiased, $\mathbb{E}[\hat{S}] = S$, so $E_L = \hat{L} - L$ is mean zero, where $\hat{L} = \hat{D} - \hat{S}$. The resulting error is missingness rather than additive noise: most pairs are left at zero and the rest rescaled by $1/p$, so it is heavy, structured, and its per-node variance grows as $p \rightarrow 0$ (Proposition 4).

The scope of the analysis is the population matrix. The empirical log-det estimator of an individual entry concentrates around S_{ij} at rate $\mathcal{O}(1/\sqrt{\ell})$ [2], and the structural conditions of Section 2 are stated as properties of the population S , whose observable proxy is that estimator.

3.3 A second operator on the same leaves

The Laplacian is not the only operator on these leaves whose spectrum carries the split. A second construction replaces S by the pairwise *distance* matrix $\mathcal{D}$ — distinct from the degree matrix D of $L = D - S$ — built entrywise from S by $\mathcal{D}_{ij} = -\log S_{ij}$, and centers it, $\mathcal{B} := H\mathcal{D}H$ with $H = I - \frac{1}{m}\mathbf{1}\mathbf{1}^\top$. There the split sits in the sign pattern of $u^{(1)}$, the eigenvector of $\mathcal{B}$ at the eigenvalue of largest absolute value, rather than in the Fiedler vector of L .

The two operators see the same tree. Under Assumptions A1 and A2 the closed form (5) gives, with d_{ij} the path length of (1),

$$\mathcal{D}_{ij} = -\log S_{ij} = |\text{tr}(Q)| d_{ij}, \quad \mathcal{D}_{ii} = 0, \quad (16)$$

so $\mathcal{D}$ is the additive path metric of $\mathcal{T}$ up to the positive scale $|\text{tr}(Q)|$, and a spectral statement about $\mathcal{B}$ is a statement about tree distances. Since (16) is entrywise, the same Bernoulli(p) mask Ω that samples S samples $\mathcal{D}$ unchanged: both operators are observed under the one model (15), and Section 4 states a sampling rate for each. ?? works out the population spectrum of $\mathcal{B}$ and proves the distance-route rate.

3.4 The recovery problem

Problem. Given the sparsely observed estimate $\hat{S}$, recover the Fiedler split $\Pi(S)$ of the full matrix from the sign pattern of the Fiedler vector $\hat{v}^{(2)}$ of $\hat{L}$. We ask for the minimal sampling rate p at which $\Pi(\hat{S}) = \Pi(S)$ with high probability. Since $\mathbb{E}[\hat{S}] = S$, the question reduces to an eigenvector-perturbation problem over $E_L = \hat{L} - L$, which the remainder of the paper analyzes.

Theorem 1 already makes $\Pi(S)$ a clan pair of $\mathcal{T}$, so a rate at which $\Pi(\hat{S}) = \Pi(S)$ is a rate at which the sub-sampled operator returns a genuine split of the tree.

Recovery is, at its core, an *entry-wise* event. Two vectors induce the same sign partition exactly when they agree in sign at every node, so the top-down partition succeeds precisely when the empirical Fiedler vector agrees in sign with its population counterpart coordinate by coordinate, that is, when

$$\left\| \hat{v}^{(2)} - v^{(2)} \right\|_{\infty} < \min_i |v_i^{(2)}|. \quad (17)$$

Condition (17) is the formal recovery event that every bound in the paper is measured against.

4 Main Result

The entry-wise recovery criterion (17) is what the sampling rate has to buy. This section converts it into a sufficient condition on p for the Laplacian $L = D - S$, stated in the structural quantities of Section 2: the margin ρ of (10) and the imbalance η of (9).

Theorem 2 (Exact recovery of the full-data split). *Let S satisfy the non-balanced CFBM of Definition 4 with m terminal nodes and imbalance $\eta \geq 1$, let ρ be the structural margin and let $\hat{v}^{(2)}$ be the empirical Fiedler vector — the unit eigenvector belonging to the second-smallest eigenvalue of the sampled Laplacian $\hat{L} = \hat{D} - \hat{S}$, formed from entries observed with probability p as in the observation model (15). Assume Condition C1. For the entry-wise recovery criterion (17) to hold with high probability, i.e. $\left\| \hat{v}^{(2)} - v^{(2)} \right\|_{\infty} < \min_i |v_i^{(2)}|$, it suffices that*

$$p > \frac{8(1+\eta)^3 \beta_0^2 \log m}{m(\rho - \eta S_{\text{out}}^{\max})^2} = \Theta\left(\frac{\log m}{m}\right). \quad (18)$$

(18) is a *sufficient* condition. No matching lower bound is proved anywhere in this paper, so the $\Theta(\log m/m)$ is the growth order of that sufficient bound and not a claim that the rate is tight or order-optimal.

The distance operator $\mathcal{B} = HDH$ of Section 3 admits a statement of the same shape — entry-wise sign agreement of an empirical structural eigenvector against a piecewise-constant target, at a rate governed by a spectral gap. Since the hypotheses of the two operators are not nested, it is a *sibling* of Theorem 2 rather than a corollary of it; ?? states it as ?? and proves it.

5 Topological Scaling and Spectral Gap

This section deals with how the macro-topology of $\mathcal{T}$ fixes its spectrum: the eigenpair that carries the primary split, the coherence of the leading invariant subspace, and the gap separating λ_2 from λ_3 . The whole dependence runs through the single structural parameter η of Definition 6.

Lemma 2 (Spectral decomposition of the imbalanced Laplacian). *Let S satisfy Definition 4 at the primary split, as Proposition 1 guarantees under Assumptions A1 and A2. Then (i) the smallest eigenvalue of L is $\lambda_1 = 0$, simple, with $v^{(1)} = \frac{1}{\sqrt{m}} \mathbf{1}_m$; (ii) $m\beta_0$ is an eigenvalue of L whose eigenvector is piecewise constant on the split, $v_i^{(2)} = +c_A$ for $i \in A$ and $v_i^{(2)} = -c_B$ for $i \in B$, with $c_A = \sqrt{b/(am)}$ and $c_B = \sqrt{a/(bm)}$; and (iii) consequently*

$$\lambda_2 \leq m\beta_0. \quad (19)$$

If in addition Condition C1 holds, then (iv) (19) holds with equality: $\lambda_2 = m\beta_0$ is simple and $v^{(2)}$ is the Fiedler vector of the similarity graph.

The proof is in Appendix A. Clauses (i)–(iii) are a direct computation, and they use only the constancy of the cross-clan block, not the floor: the identity $(D_A - S_A)\mathbf{1}_a = b\beta_0\mathbf{1}_a$ holds for an arbitrary symmetric diagonal block, because D_A already carries the row sums of that block. Clause (iv) is where Condition C1 enters, as the statement that $m_{\min}S_{\text{in}}^{\min} > m\beta_0$: it puts the interlacing lower bound on λ_3 used in Lemma 3 strictly above $m\beta_0$, leaving no eigenvalue between 0 and $m\beta_0$. It is what makes $\min_i |v_i^{(2)}| = c_B$ available to the proof of Theorem 2.

5.1 Coherence and the spectral gap under imbalance

Two scalar quantities of the population Laplacian sit behind the constants of (18), and η moves both against recovery. The first is the spectral gap that isolates the Fiedler vector,

$$\Delta\lambda := \lambda_3 - \lambda_2, \quad (20)$$

with λ_2, λ_3 as in Definition 5. The second measures how evenly the leading invariant subspace of L spreads across nodes.

Definition 8 (Subspace coherence). Let $U = [v^{(1)}, v^{(2)}] \in \mathbb{R}^{m \times 2}$ span the invariant subspace of the two smallest eigenvalues of L , with $v^{(1)} = \frac{1}{\sqrt{m}}\mathbf{1}_m$ the trivial flat eigenvector and $v^{(2)}$ the Fiedler vector. The *subspace coherence* is

$$\mu(U) := \frac{m}{2} \max_{1 \leq i \leq m} \|P_U e_i\|_2^2 = \frac{m}{2} \max_{1 \leq i \leq m} ((v_i^{(1)})^2 + (v_i^{(2)})^2), \quad P_U = UU^\top. \quad (21)$$

Coherence is minimal, $\mu(U) = 1$, when the subspace is spread evenly, and grows as it localizes onto few coordinates. Both quantities are fixed by macro-topological features alone: η first inflates $\mu(U)$, then erodes $\Delta\lambda$.

Proposition 2 (Topological scaling of coherence). *Under Definition 4, the tree imbalance η maps deterministically to the subspace coherence,*

$$\mu(U) = \frac{1 + \eta}{2}. \quad (22)$$

The identity follows by substituting the Fiedler coordinates of Lemma 2 into (21); the short computation is given in Appendix A. Thus a balanced split attains the optimal coherence $\mu(U) = 1$, and the signal spreads more unevenly across nodes as η grows; the generated regime of Section 7 confirms the identity on data from the full generative pipeline. The identity is reported as a structural diagnostic: since the noise bound of Lemma 4 is obtained from the row norms of S directly (Appendix B), $\mu(U)$ does not itself enter the sampling rate, and the η -dependence of (18) arrives through the margin and the gap instead.

The gap $\Delta\lambda$ of (20) is the quantity the perturbation argument of Section 6 divides by, and the same imbalance that raises $\mu(U)$ drives it toward zero.

Lemma 3 (Spectral-gap lower bound). *Let L be the combinatorial Laplacian of a similarity matrix satisfying the molecular clock, with structural margin $\rho = S_{\text{in}}^{\min} - S_{\text{out}}^{\max}$. For $m_{\min} = \min(a, b)$ and imbalance η , the spectral gap isolating the Fiedler vector satisfies*

$$\Delta\lambda \geq m_{\min}\rho - \eta m_{\min} S_{\text{out}}^{\max}. \quad (23)$$

The full algebraic-connectivity derivation is given in Appendix A, building on the eigenpairs of Lemma 2. The bound (23) is purely a CFBM statement, controlling the gap through the abstract minimum intra-clan similarity $S_{\text{in}}^{\min}$ of (6).

5.2 Two failure modes

Proposition 2 and Lemma 3 show that the same parameter η acts on both spectral quantities at once, which produces two distinct ways for top-down recovery to fail; together they bound the admissible region of (m, η) . The first is *structural collapse*, $\Delta\lambda \rightarrow 0$, which we state here as a corollary of the gap lemma; the second is *statistical infeasibility*, $p > 1$, which is a consequence of Theorem 2. A positive gap requires the structural margin to dominate the imbalance-weighted cross-clan boundary, $m_{\min}\rho > \eta m_{\min} S_{\text{out}}^{\max}$, i.e. $\eta < S_{\text{in}}^{\min}/S_{\text{out}}^{\max}$ — precisely what Condition C1 records. Translating this ratio into tree distances bounds the imbalance the topology can sustain.

Corollary 1 (Structural imbalance tolerance). *Write similarities as distances, $S_{ij} = e^{-d(x_i, x_j)}$, and let $r(\mathcal{T}) = \max_{i,j}(-\log S_{ij})$ be the diameter and $h(\mathcal{T})$ the depth of $\mathcal{T}$ [2], with $h(\mathcal{T}) < r(\mathcal{T})$. Then a positive spectral gap is possible only if*

$$\eta \leq \mathcal{O}(e^{\mathcal{O}(h(\mathcal{T})) - r(\mathcal{T})}). \quad (24)$$

In particular, for an average Kingman coalescent tree, where $r(\mathcal{T}), h(\mathcal{T}) = \mathcal{O}(\log m)$ [14, 3], this gives $\eta \leq \mathcal{O}(m^c)$ for a model-dependent constant c .

The proof sketch (substituting the diameter/depth bounds into $\eta < S_{\text{in}}^{\min}/S_{\text{out}}^{\max}$) is given in Appendix A.

So under balanced generative models the gap survives polynomial imbalance. A positive gap is necessary but not sufficient: even when $\Delta\lambda > 0$, the rate (18) grows with η , and the second failure mode is the imbalance past which that demand exceeds $p = 1$.

Corollary 2 (Vacuity of the certificate under imbalance). *Under the hypotheses of Theorem 2, with β_0 and the margin $\rho - \eta S_{\text{out}}^{\max}$ held bounded away from 0 independently of η , the right-hand side of (18) exceeds 1 once $\eta = \Omega(\sqrt[3]{m/\log m})$; in general the threshold is reached when $\eta^3 \beta_0^2 / (\rho - \eta S_{\text{out}}^{\max})^2 = \Omega(m/\log m)$. No admissible sampling rate $p \in (0, 1]$ then satisfies the sufficient condition, so Theorem 2 certifies nothing in that regime. What the theorem stops certifying there is the sampling: Theorem 1 continues to make $\Pi(S)$ a clan pair of $\mathcal{T}$ whatever η does.*

6 Proof Outline

The proof of Theorem 2 compares two quantities evaluated at the same node: the entry-wise error $|\hat{v}_i^{(2)} - v_i^{(2)}|$ and the entry-wise margin $|v_i^{(2)}|$. It runs in three stages — population geometry fixes the margin and noise budget, a resolvent expansion reduces the error at each node to a scalar, and a per-node Bernstein bound controls that scalar — after which imposing (17) yields the rate (18). Each stage is stated as a lemma; the proofs are in Appendices A to C.

Stage 1 (population geometry and noise budget). The same imbalance η sets both sides of (17) at once — it fixes the margin and the spectral gap through the piecewise-constant Fiedler vector of Lemma 2 and the gap bound of Lemma 3, kept positive by Condition C1, while on the other side matrix Bernstein applied to the sampling mask bounds the noise the gap has to absorb.

Lemma 4 (Laplacian spectral error via matrix Bernstein). *Under Definition 4 and the observation model (15), there are constants $\tilde{C}, C_0$ such that if $p \geq C_0 \frac{\log m}{m}$ then with high probability*

$$\|E_L\|_2 \leq \tilde{C} \sqrt{\frac{m \log m}{p}}.$$

The proof is in Appendix B. Compared against the gap bound of Lemma 3, this is what certifies the strong-signal regime $\|E_L\|_2 < \Delta\lambda$ that the next stage assumes. Under imbalance the margin and the gap are extremal at different nodes, which is what makes the final synthesis η -dependent.

Stage 2 (entry-wise reduction). Since (17) is an ℓ_∞ event while Davis–Kahan controls only ℓ_2 , and the sole dimension-free transfer between them, $\|\cdot\|_\infty \leq \|\cdot\|_2$, charges every node as though it were the worst one, the reduction has to be coordinate-wise: in the regime of Stage 1 a resolvent (Neumann) expansion of $\hat{v}^{(2)}$ in E_L trades the operator norm for a single scalar per coordinate, at the price of a remainder that is second order in the noise-to-gap ratio.

Lemma 5 (Neumann decomposition of the Fiedler vector). *Let L be symmetric with eigenpairs $\{(\lambda_\ell, v^{(\ell)})\}$, let $\hat{L} = L + E_L$ be a symmetric perturbation, and write $\hat{v}^{(2)} = v^{(2)} + w$. If $\|E_L\|_2 < \Delta\lambda$, then, with Π the spectral projector onto the complement of $v^{(2)}$, w admits the exact resolvent expansion $w = \sum_{k \geq 1} (-1)^k (\Delta\lambda^{-1} \Pi E_L)^k v^{(2)}$, whose first-order term is*

$$w^{(1)} = \sum_{\ell \neq 2} \frac{-\langle v^{(\ell)}, E_L v^{(2)} \rangle}{\lambda_\ell - \hat{\lambda}_2} v^{(\ell)}, \quad (25)$$

so that entrywise $w_i = -(E_L v^{(2)})_i / \Delta\lambda + R_i$ with the remainder of second order, $\|R\|_\infty = O(\|E_L\|_2^2 / \Delta\lambda^2)$.

The expansion is due to Eldridge et al. [13], Fan et al. [15] and Abbe et al. [1] rather than to this paper, which is why it is stated as a lemma; the decomposition and its first-order bound are proved in Appendix C.1, and the remainder R separately in Appendix C.3 (Lemma 10), its discard being what confines the argument to the strong-signal regime of Stage 1.

Stage 3 (per-node concentration). The surviving scalar $(E_L v^{(2)})_i$ is a sum of independent, mean-zero increments, each carrying the factor $v_i^{(2)} - v_j^{(2)}$. Every within-clan increment therefore vanishes *exactly* — this is a cancellation, not a bound, and it rests on the clock (Assumption A2) making $v^{(2)}$ exactly piecewise constant, so it is the one step that would degrade if the cross-clan block were only approximately constant. Only cross-clan pairs survive, Bernstein’s inequality applies node by node, and since this is the only place the sampling rate p enters the final condition, we state the bound in the main text.

Lemma 6 (Per-node concentration). *With scale $R_B = \beta_0(1 + \eta)/(p\sqrt{m\eta})$, for each $i \in B$ and $\delta > 0$, with probability $\geq 1 - \delta$, $|(E_L v^{(2)})_i| \leq 2\sqrt{2\sigma_B^2 \ln(2/\delta)} + \frac{2}{3}R_B \ln(2/\delta)$. In the variance-dominated regime ($p = \Theta(\log m/m)$), a union bound over all m nodes at $\delta = \delta_0/m$ gives, with high probability uniformly over i ,*

$$|(E_L v^{(2)})_i| \lesssim \sqrt{\frac{8(1 + \eta)\beta_0^2 \log(2m/\delta_0)}{\eta p}}. \quad (26)$$

The increment representation, the cancellation of the within-clan pairs, the per-node variance σ_B^2 and the proof are in Appendix C.2. This is the step that realises the $\sqrt{m}$ saving that the Davis–Kahan route forfeits: the variance of the increment sum is smaller than $\|E_L\|_2^2$ by a factor m , and it is this per-coordinate saving that turns (17) into the rate (18); Appendix D runs the two routes side by side and Remark 2 reports the ratio. The three stages now combine.

Proof of Theorem 2. Step 1 (population geometry). Lemmas 2 and 3 supply the right-hand side of (17) as $\min_i |v_i^{(2)}| = c_B$ and a spectral gap $\Delta\lambda$ that is positive under Condition C1, and Lemma 4 places $\|E_L\|_2$ below it.

Step 2 (entry-wise reduction). Lemma 5 is therefore applicable and gives $w_i = -(E_L v^{(2)})_i / \Delta\lambda + R_i$; discarding R as lower order against c_B reduces (17) to the per-node condition $|(E_L v^{(2)})_i| < \Delta\lambda |v_i^{(2)}|$ holding at every node i .

Step 3 (concentration). Lemma 6 bounds the left-hand side of that condition uniformly over the m nodes.

Step 4 (synthesis). The per-node condition is tightest on clan B , since $\eta \geq 1$. Squaring $\sqrt{8\sigma_B^2 \log m} < \Delta\lambda c_B$ and substituting $\sigma_B^2, c_B^2 = 1/(m\eta)$ and the gap bound $\Delta\lambda \geq m_{\min}(\rho - \eta S_{\text{out}}^{\max})$, the factor η cancels and solving for p gives exactly (18) (Proposition 5 and Corollary 4). Under (18) every node satisfies $|w_i| < |v_i^{(2)}|$, hence $\hat{v}_i^{(2)}$ and $v_i^{(2)}$ share a sign; since $v^{(2)}$ is sign-constant on each clan (Lemma 2), the zero-threshold partition of $\hat{v}^{(2)}$ is (A, B) exactly. $\square$

?? states the counterpart of Theorem 2 for the centered distance operator. Its proof follows the same four steps, with the population geometry of Step 1 and the variance load of Step 3 replaced by their distance analogues; Lemma 5 is applied unchanged, being a statement about an arbitrary symmetric perturbation of an arbitrary symmetric matrix. ?? carries the details.

7 Empirical Results

Two experiments produce the figures below. In the first, S is written down directly from the constant-block CFBM of Definition 4 with $S_{\text{in}}^{\min} = 0.9$ and $S_{\text{out}}^{\max} = 0.05$, so the sampling mask is the only source of error. In the second, dendropy simulates Kingman coalescent trees, evolves JC69 sequences of length $\ell = 10^4$ along their branches, and S is the log-det similarity estimated from the resulting alignment (Assumption A1); tree shape is then off-model and every entry of S carries estimation noise, so this is where the exact cross-clan baseline of Assumption A2 and the margin of Condition C1 are put to the test.

Each matrix is sub-sampled independently at every p on a grid and read through both operators of Section 3 — the Laplacian L of Theorem 2, rounded by k -means on L_{sym} , and the centred distance operator $\mathcal{B} = HDH$ of ??, rounded by the sign of $u^{(1)}$ — each as its own theorem prescribes. A point is an average over independent sub-sampling replicates at that size and imbalance, 20 per point on the similarity route and 8 on the distance route; on the simulated trees the count is set by how many trees the pool holds at that (n, η) .

Recovery is scored by the Normalized Mutual Information (NMI) between the split $\hat{C} = \Pi(\hat{S})$ returned from the sub-sampled matrix and the reference split $C^* = \Pi(S)$ returned by the same operator on the full matrix at $p = 1$. NMI is the standard measure of agreement between two spectral-clustering partitions [17, 23]:

$$\text{NMI}(Y, C) = \frac{2I(Y; C)}{H(Y) + H(C)}, \quad (27)$$

with $I(Y; C)$ the mutual information and $H(\cdot)$ the Shannon entropy. NMI attains its maximum value 1 if and only if the two partitions are identical up to relabeling, and decays as they disagree [28, 17]. The quantity being measured is therefore the cost of sub-sampling and nothing else: both partitions come from the same operator on the same tree, and they differ only in how many entries of S were computed.

Corollary 3 (Perfect partition recovery and $\text{NMI} = 1$). *Under the hypotheses of Theorem 2, the partition $\hat{C}$ obtained by thresholding $\hat{v}^{(2)}$ at zero satisfies $\hat{C} = C^*$ with high probability, and therefore $\text{NMI}(\hat{C}, C^*) = 1$.*

Proof. Theorem 2 guarantees $\|\hat{v}^{(2)} - v^{(2)}\|_\infty < \min_i |v_i^{(2)}|$, so for each node i , $|\hat{v}_i^{(2)} - v_i^{(2)}| < |v_i^{(2)}|$, which forces $\hat{v}_i^{(2)}$ and $v_i^{(2)}$ to share a sign. Thresholding both at zero therefore yields the same partition, $\hat{C} = \Pi(\hat{S}) = \Pi(S) = C^*$. No property of the tree enters: the argument is node-wise sign agreement between two vectors. Exact agreement gives $I(\hat{C}; C^*) = H(\hat{C}) = H(C^*)$, whence by (27) $\text{NMI}(\hat{C}, C^*) = 1$. $\square$

All four figures below share one layout: one pair of panels per imbalance, at $\eta = 1, 5, 10, 15$ throughout. On the left, measured NMI against the sampling rate p , one curve per size, the curves running dark to light (viridis) with increasing size, and the red dashed line at the 0.90 level that defines the empirical threshold $\hat{p}^*$. On the right, $\hat{p}^*$ against size on log-log axes, with a dotted reference of the shape the governing rate predicts — $\log m/m$, from (18) on the similarity side and (??) on the distance side — fitted to the measured points. Filled markers are measured; open markers are bounds rather than thresholds, a read-off pinned at either end of the p grid, and are excluded from the fit.

Along the similarity route the transition sharpens and shifts left as the size grows, and $\hat{p}^*$ decays parallel to the reference at every η on both data sets: on the closed-form matrix of Figure 3, and on the simulated phylogenies of Figure 4, where the coherence and gap identities of Proposition 2 and Lemma 3 continue to hold despite off-model tree shapes and JC69 noise. The $\Theta(\log m/m)$ sample complexity of (18) therefore survives finite-sequence estimation noise.

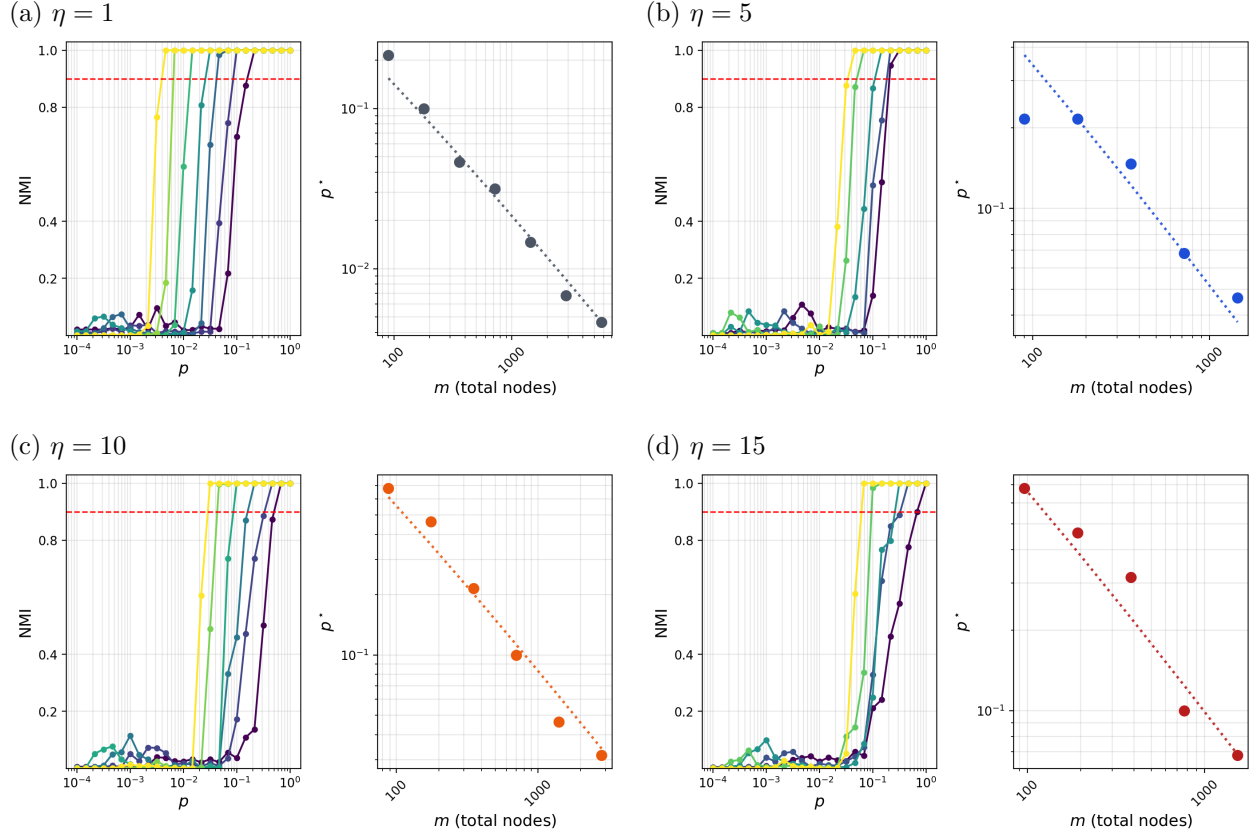


Figure 3: **Similarity route, no estimation noise.** S is the constant-block CFBM in closed form, so the sampling mask is the only error present. The transition sharpens and moves left as m grows, and $\hat{p}^*$ falls parallel to $\log m/m$ at every imbalance.

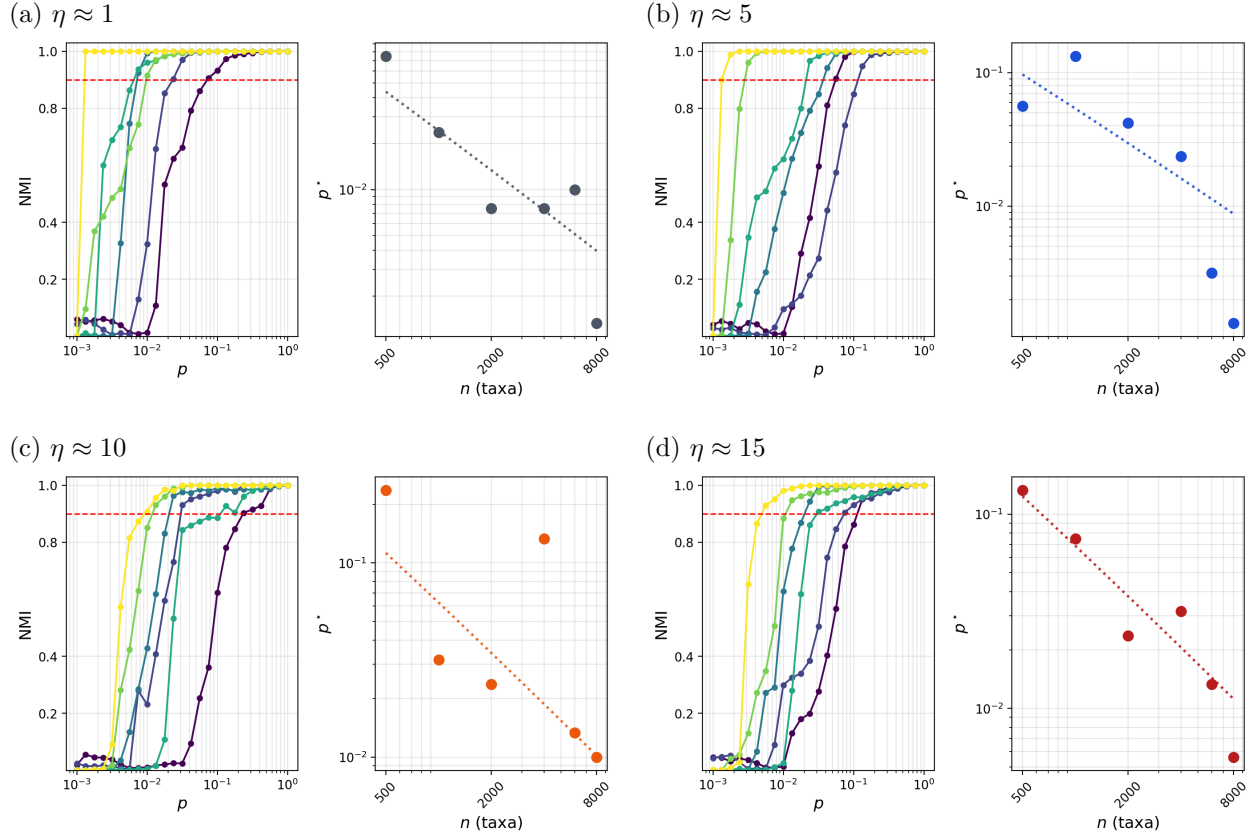


Figure 4: **Similarity route, simulated phylogenies.** Kingman trees carrying JC69 sequences, with S estimated from the alignment rather than written down. The $\log m/m$ decay of Figure 3 survives off-model tree shapes and per-entry estimation noise.

Run through $\mathcal{B}$ on the same data, the sign rule of ?? reproduces that decay at $\eta = 1$ and degrades progressively away from balance, until at $\eta = 15$ most synthesized sizes never recover below full data (Figures 5 and 6). Only the $\eta = 1$ panel of Figure 5 lies inside the configuration where hypotheses (i)–(ii) of ?? are verified (??); the remaining panels measure the estimator rather than certify the rate. Which operator tolerates more noise is not read off these sweeps — the two routes differ in rounding rule as well as in operator — and ?? compares the two rates’ parameter groups, not the two estimators.

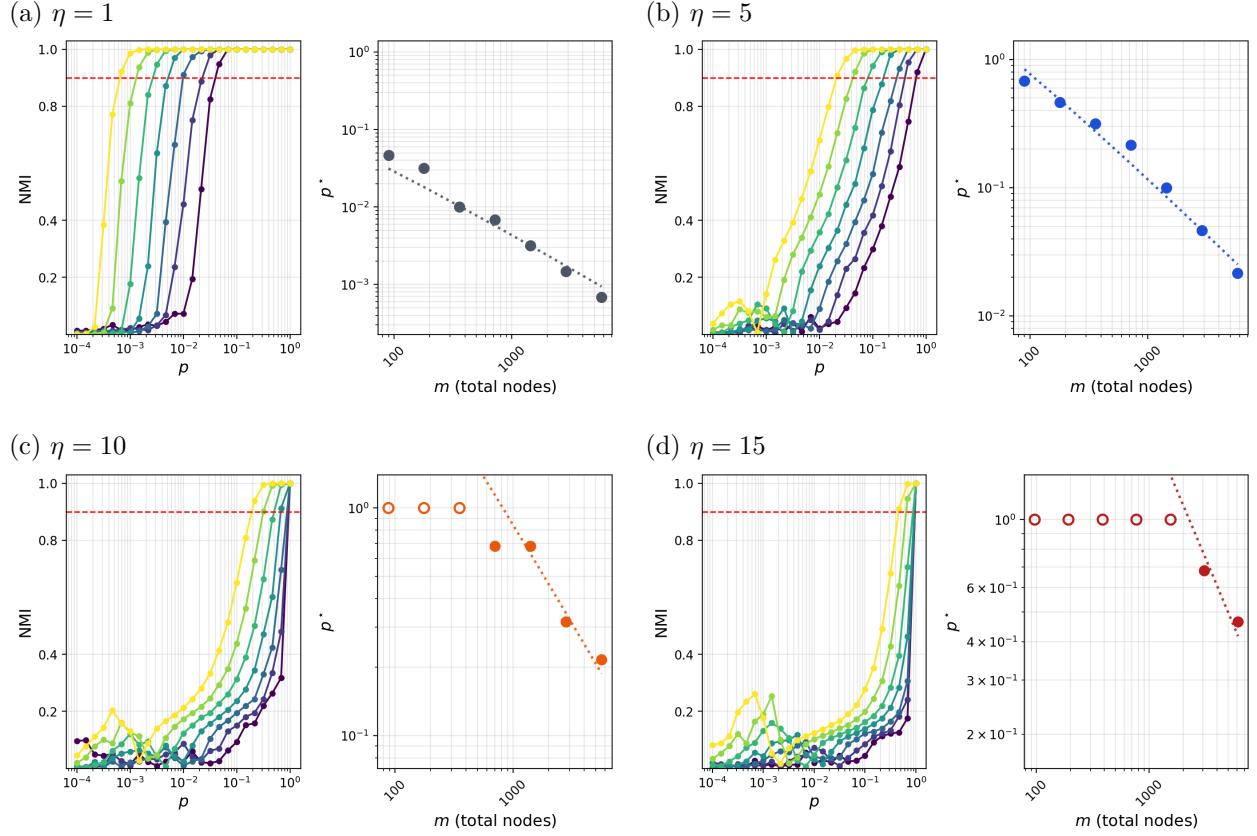


Figure 5: **Distance route, no estimation noise.** The matrix of Figure 3 read through $\mathcal{B} = HDH$ with $\mathcal{D} = -\log S$. The decay is reproduced at $\eta = 1$ and degrades as imbalance grows, until at $\eta = 15$ most sizes reach the threshold only at full data.

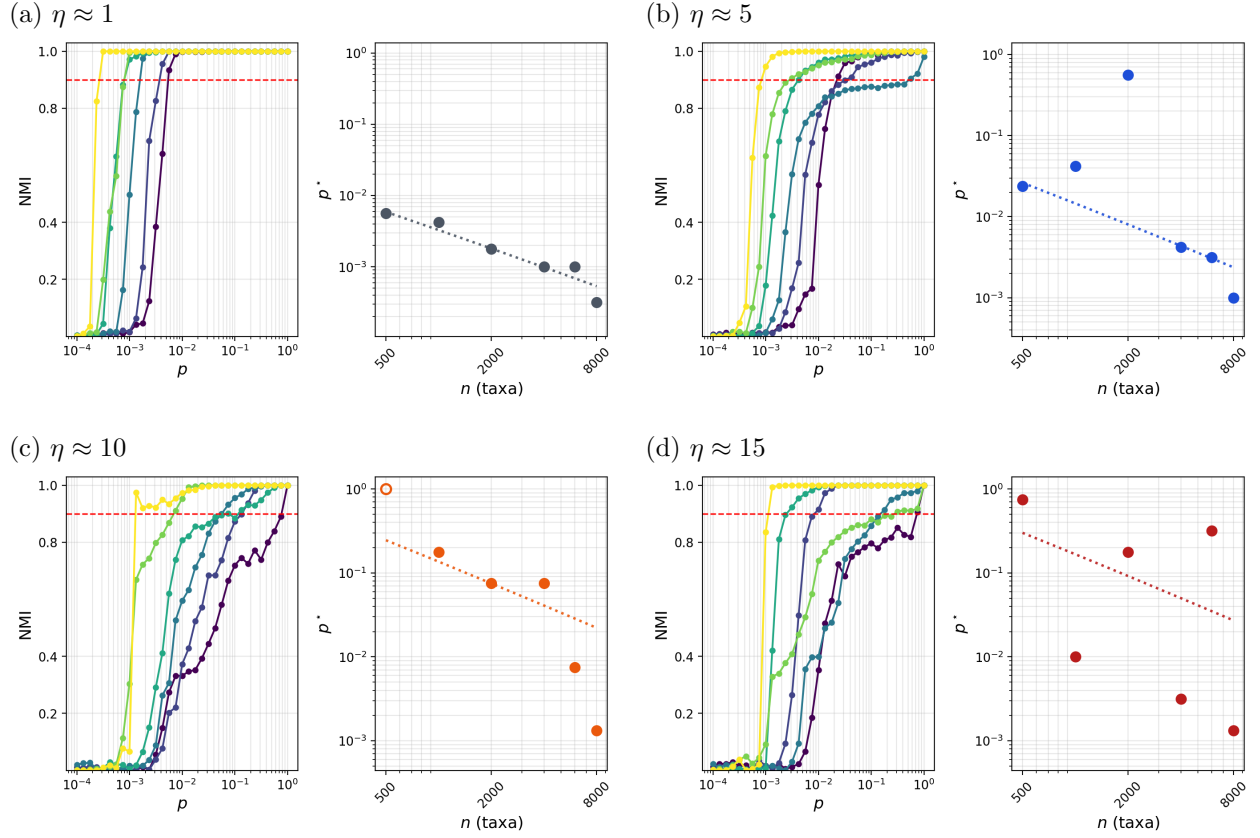


Figure 6: **Distance route, simulated phylogenies.** The trees, sequences and p grid of Figure 4, read through $\mathcal{B}$ instead of L . Estimation noise leaves the picture of Figure 5 unchanged: the balanced case decays, and imbalance costs the threshold.

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A Proof of the spectral geometry

Every later appeal to a bounded population entry is to Proposition 1. No range claim is made for $\hat{S}$: by (15) its observed off-diagonal entries equal S_{ij}/p , which exceed 1 throughout the regime $p = \Theta(\log m/m)$ of interest.

We first prove the spectral decomposition stated in Section 2, which the proof of Lemma 3 then uses.

Proof of Lemma 2(i)–(iii). For any $L = D - S$, $L\mathbf{1} = D\mathbf{1} - S\mathbf{1} = 0$, so 0 is an eigenvalue with $v^{(1)} = \frac{1}{\sqrt{m}}\mathbf{1}_m$; no assumption on S is needed. Every off-diagonal entry of S is at least $\beta_0 > 0$ by Proposition 1, so the similarity graph is connected and $\lambda_1 = 0$ is simple.

Step heights. Assume $v^{(2)}$ has the stated form with $c_A, c_B > 0$. The constraints $\|v^{(2)}\|_2 = 1$ and $\langle v^{(1)}, v^{(2)} \rangle = 0$ give $ac_A = bc_B$, hence

$$a\left(\frac{b}{a}c_B\right)^2 + bc_B^2 = 1 \implies \frac{bm}{a}c_B^2 = 1 \implies c_B^2 = \frac{a}{bm}, \quad c_A^2 = \frac{b}{am}.$$

Eigenpair verification. Write $v^{(2)} = (c_A\mathbf{1}_a, -c_B\mathbf{1}_b)^\top$ with $ac_A = bc_B$. The only structural input is that the diagonal sub-degrees of Proposition 1 already carry the row sums of their own block, $D_A = \text{diag}(S_A\mathbf{1}_a) + b\beta_0 I_a$, so that

$$(D_A - S_A)\mathbf{1}_a = b\beta_0\mathbf{1}_a, \quad (D_B - S_B)\mathbf{1}_b = a\beta_0\mathbf{1}_b, \quad (28)$$

identically, for arbitrary symmetric S_A, S_B : no constancy of the diagonal blocks is used, and the intra-clan floor of (6) plays no role here. Using the block form of L with $J_{a \times b}\mathbf{1}_b = b\mathbf{1}_a$, $J_{b \times a}\mathbf{1}_a = a\mathbf{1}_b$ and (28),

$$Lv^{(2)} = \begin{pmatrix} b\beta_0 c_A \mathbf{1}_a + b\beta_0 c_B \mathbf{1}_a \\ -a\beta_0 c_A \mathbf{1}_b - a\beta_0 c_B \mathbf{1}_b \end{pmatrix},$$

which reduces to $(b\beta_0(c_A + c_B)\mathbf{1}_a, -a\beta_0(c_A + c_B)\mathbf{1}_b)^\top$. From $ac_A = bc_B$, $b(c_A + c_B) = mc_A$ and $a(c_A + c_B) = mc_B$, so $Lv^{(2)} = m\beta_0 v^{(2)}$: the pair $(m\beta_0, v^{(2)})$ is an eigenpair, which is claim (ii).

The bound (19). Since $\lambda_1 = 0$ is simple with eigenvector $v^{(1)}$, the Courant–Fischer characterisation of the second eigenvalue reads $\lambda_2 = \min\{x^\top Lx : \|x\|_2 = 1, x \perp v^{(1)}\}$. The vector $v^{(2)}$ is admissible in that minimisation — it is a unit vector and $\langle v^{(1)}, v^{(2)} \rangle = 0$ by construction — so $\lambda_2 \leq \langle v^{(2)}, Lv^{(2)} \rangle = m\beta_0 \|v^{(2)}\|_2^2 = m\beta_0$. $\square$

Nothing above forces λ_2 to *equal* $m\beta_0$; that is an ordering claim, and it is what Condition C1 supplies in clause (iv), proved at the end of this appendix after the interlacing bound it needs.

Proof of Proposition 2. Substituting the Fiedler coordinates of Lemma 2 into (21),

$$\mu(U) = \frac{m}{2} \max\left\{\frac{1}{m} + c_A^2, \frac{1}{m} + c_B^2\right\} = \frac{m}{2} \left(\frac{1}{m} + \frac{b}{ma}\right) = \frac{1}{2} \left(1 + \frac{b}{a}\right) = \frac{1+\eta}{2}. \quad \square$$

Proof of Lemma 3. The argument uses Assumption A2 and Proposition 1, and only the *inequality* (19), $\lambda_2 \leq m\beta_0 = mS_{\text{out}}^{\max}$ — not clause (iv) of Lemma 2, whose proof appeals to the interlacing step below. For λ_3 , standard algebraic-connectivity interlacing gives $\lambda_3(L) \geq \min(\lambda_2(L_{C_1}), \lambda_2(L_{C_2}))$, and applying the Fiedler-eigenvalue formula to each sub-clan Laplacian gives $\lambda_2(L_{C_c}) \geq n_c S_{\text{in}}^{(c)}$. Since $\min_c S_{\text{in}}^{(c)} = S_{\text{in}}^{\min}$, the weakest intra-clan similarity of (6),

$$\lambda_3 \geq \min(\lambda_2(L_{C_1}), \lambda_2(L_{C_2})) \geq \min(n_1, n_2) S_{\text{in}}^{\min} = m_{\min} S_{\text{in}}^{\min}, \quad (29)$$

so $\Delta\lambda = \lambda_3 - \lambda_2 \geq m_{\min} S_{\text{in}}^{\min} - m S_{\text{out}}^{\max}$. Using $S_{\text{in}}^{\min} = \rho + S_{\text{out}}^{\max}$, which is (11), and $m = m_{\min}(1 + \eta)$,

$$\Delta\lambda \geq m_{\min}(\rho + S_{\text{out}}^{\max}) - (1 + \eta)m_{\min} S_{\text{out}}^{\max} = m_{\min}\rho - \eta m_{\min} S_{\text{out}}^{\max}. \quad \square$$

Proof of Corollary 1. Positivity of the bound in Lemma 3 requires $\eta < S_{\text{in}}^{\min}/S_{\text{out}}^{\max}$. The minimum intra-clan similarity is bounded by the diameter, $S_{\text{in}}^{\min} \geq e^{-r(\mathcal{T})}$, and the maximum cross-clan similarity by the subtree depth, $S_{\text{out}}^{\max} \propto e^{-\mathcal{O}(h(\mathcal{T}))}$. Substituting into $\eta < S_{\text{in}}^{\min}/S_{\text{out}}^{\max}$ gives (24); the logarithmic scaling $r, h = \mathcal{O}(\log m)$ under Kingman yields $\eta \leq \mathcal{O}(e^{\mathcal{O}(\log m)}) = \mathcal{O}(m^c)$. $\square$

With (29) in hand the inequality (19) can be closed.

Proof of Lemma 2(iv). Suppose $\lambda_2 < m\beta_0$. Then $m\beta_0$, an eigenvalue of L by Lemma 2(ii) and equal to neither $\lambda_1 = 0$ nor λ_2 , satisfies $m\beta_0 \geq \lambda_3$. The interlacing bound (29) gives $\lambda_3 \geq m_{\min} S_{\text{in}}^{\min}$, so this forces $m_{\min} S_{\text{in}}^{\min} \leq m\beta_0$. But $S_{\text{out}}^{\max} = \beta_0$ under Assumption A2 and $m = m_{\min}(1 + \eta)$, so (13) reads $S_{\text{in}}^{\min} > (1 + \eta)\beta_0$, i.e. $m_{\min} S_{\text{in}}^{\min} > m\beta_0$ — a contradiction. Hence $\lambda_2 = m\beta_0$, and then $\lambda_3 \geq m_{\min} S_{\text{in}}^{\min} > m\beta_0 = \lambda_2$ shows λ_2 is simple, so $v^{(2)}$ is the Fiedler vector up to sign. $\square$

B Proof of the Laplacian sampling error

We prove Lemma 4 by bounding $\|E_L\|_2 \leq \|E_D\|_2 + \|E_S\|_2$, where $E_S = \hat{S} - S$ and $E_D = \hat{D} - D$. The only structural fact needed is the entry range of Proposition 1. By (15) the diagonal is observed exactly, $\hat{S}_{ii} = S_{ii} = 1$, so it carries no noise and only off-diagonal pairs enter the sums below; on those $S_{ij} \leq S_{\max} < 1$, giving

$$\max_i \sum_{j \neq i} S_{ij}^2 \leq (m - 1) S_{\max}^2 < m S_{\max}^2. \quad (30)$$

Proof of Lemma 4. Bounding $\|E_S\|_2$. Write $E_S = \sum_{i < j} Z^{(ij)}$, where $Z^{(ij)}$ is supported on the symmetric off-diagonal pair (i, j) with value $(\frac{1}{p}\Omega_{ij} - 1)S_{ij}$; the summands are independent and mean zero. The scale is $\|Z^{(ij)}\|_2 \leq \max(\frac{1}{p} - 1, 1)S_{ij} \leq S_{\max}/p =: R$. For the variance proxy, $\mathbb{E}[(Z_{ij}^{(ij)})^2] = \frac{p(1-p)}{p^2} S_{ij}^2 \leq \frac{1}{p} S_{ij}^2$, so by (30),

$$\sigma^2 \leq \frac{1}{p} \max_i \sum_{j \neq i} S_{ij}^2 \leq \frac{1}{p} m S_{\max}^2.$$

Matrix Bernstein [29], $\mathbb{P}(\|E_S\|_2 \geq t) \leq 2m \exp(\frac{-t^2/2}{\sigma^2 + Rt/3})$, gives in the variance-dominated regime, with probability $1 - \epsilon$, $\|E_S\|_2 \leq 2S_{\max} \sqrt{m \ln(2m/\epsilon)/p}$.

Bounding $\|E_D\|_2$. E_D is diagonal with $(E_D)_{ii} = \sum_{j \neq i} (\frac{1}{p}\Omega_{ij} - 1)S_{ij}$; a scalar Bernstein with $\sigma_D^2 \leq \frac{1}{p} m S_{\max}^2$ and $R_D = S_{\max}/p$ gives $\|E_D\|_2 \lesssim S_{\max} \sqrt{m \ln(2m/\epsilon)/p}$.

Synthesis. The two terms are now of the same order, so

$$\|E_L\|_2 \leq \|E_S\|_2 + \|E_D\|_2 \leq \tilde{C} \sqrt{\frac{m \log m}{p}},$$

with $\tilde{C}$ absorbing $S_{\max}$ and absolute constants. The bound is uniform in η : the imbalance enters the argument only through the gap $\Delta\lambda$ this noise is compared against, not through the noise itself. $\square$

C Proof of the entry-wise perturbation

C.1 Neumann decomposition (proof of Lemma 5)

Write $\hat{L} = L + E_L$ and $\hat{v}^{(2)} = v^{(2)} + w$, and expand w into a resolvent power series in E_L [13].

Lemma 7 (Exact Neumann decomposition). *Define*

$$w^{(1)} = \sum_{\ell \neq 2} \frac{-\langle v^{(\ell)}, E_L v^{(2)} \rangle}{\lambda_\ell - \hat{\lambda}_2} v^{(\ell)}, \quad w^{(2)} = \sum_{\ell \neq 2} \frac{-\langle v^{(\ell)}, E_L w \rangle}{\lambda_\ell - \hat{\lambda}_2} v^{(\ell)} + \langle v^{(2)}, w \rangle v^{(2)}.$$

Then $w = w^{(1)} + w^{(2)}$ exactly, and under $\|E_L\|_2 < \Delta\lambda$ this is the resolvent series $w = \sum_{k \geq 1} (-1)^k (\Delta\lambda^{-1} \Pi E_L)^k v^{(2)}$.

Proof. From $\hat{L}\hat{v}^{(2)} = \hat{\lambda}_2 \hat{v}^{(2)}$ and $Lv^{(2)} = \lambda_2 v^{(2)}$, $(L - \hat{\lambda}_2 I)w = (\hat{\lambda}_2 - \lambda_2)v^{(2)} - E_L v^{(2)} - E_L w$. Projecting onto $v^{(\ell)}$ ($\ell \neq 2$), using $\langle v^{(\ell)}, v^{(2)} \rangle = 0$, gives $\langle v^{(\ell)}, w \rangle = (-\langle v^{(\ell)}, E_L v^{(2)} \rangle - \langle v^{(\ell)}, E_L w \rangle) / (\lambda_\ell - \hat{\lambda}_2)$. Expanding $w = \sum_{\ell} \langle v^{(\ell)}, w \rangle v^{(\ell)}$ and separating the $\ell = 2$ component recovers the decomposition; iterating on $E_L w$ under $\|E_L\|_2 < \Delta\lambda$ yields the convergent series. $\square$

Lemma 8 (Entry-level first-order bound). *Provided $\|E_L\|_2 < \Delta\lambda$, for every node i ,*

$$|w_i^{(1)}| \leq \frac{|(E_L v^{(2)})_i| + \|E_L\|_2 |v_i^{(2)}|}{\Delta\lambda - \|E_L\|_2}. \quad (31)$$

Proof. By Weyl's inequality $|\lambda_\ell - \hat{\lambda}_2| \geq \Delta\lambda - \|E_L\|_2$ for $\ell \neq 2$. Factoring the denominator, $|w_i^{(1)}| \leq |(P_\perp E_L v^{(2)})_i| / (\Delta\lambda - \|E_L\|_2)$ with $P_\perp = I - v^{(2)}(v^{(2)})^\top$. Since $P_\perp x = x - \langle v^{(2)}, x \rangle v^{(2)}$ and $|\langle v^{(2)}, E_L v^{(2)} \rangle| \leq \|E_L\|_2 \|v^{(2)}\|_2^2 = \|E_L\|_2$, $|(P_\perp E_L v^{(2)})_i| \leq |(E_L v^{(2)})_i| + \|E_L\|_2 |v_i^{(2)}|$. $\square$

Remark 1 (Asymmetric correction in the non-balanced CFBM). The correction $\|E_L\|_2 |v_i^{(2)}|$ equals $\|E_L\|_2 c_A$ for $i \in A$ and $\|E_L\|_2 c_B$ for $i \in B$. For the binding clan B this is $\|E_L\|_2 / \sqrt{m\eta}$, strictly sub-dominant to the recovery target c_B when $\|E_L\|_2 \ll \Delta\lambda$. Throughout Appendix C.2 we therefore use the leading-order simplification $|w_i^{(1)}| \lesssim |(E_L v^{(2)})_i| / \Delta\lambda$ for $i \in B$.

C.2 Per-node Bernstein concentration

Under the leading-order simplification of Remark 1 the entry-wise error at node i is governed by the scalar $(E_L v^{(2)})_i$, which we now represent as a sum of independent increments and concentrate.

Lemma 9 (Laplacian increment representation). *For every node i , $(E_L v^{(2)})_i = \sum_{j \neq i} X_{ij}$ with $X_{ij} = (\frac{\omega_{ij}}{p} - 1)S_{ij}(v_i^{(2)} - v_j^{(2)})$ independent and mean zero.*

Proof. With $(E_S)_{ij} = (\omega_{ij}/p - 1)S_{ij}$, $(E_L)_{ii} = \sum_{k \neq i} (E_S)_{ik}$ and $(E_L)_{ij} = -(E_S)_{ij}$, so $(E_L v^{(2)})_i = \sum_{k \neq i} (E_S)_{ik} v_i^{(2)} - \sum_{j \neq i} (E_S)_{ij} v_j^{(2)} = \sum_{j \neq i} (E_S)_{ij} (v_i^{(2)} - v_j^{(2)})$. $\square$

Proposition 3 (Within-clan terms vanish). *Under Definition 4, $v^{(2)}$ is constant on each clan (Lemma 2), so $X_{ij} = 0$ for same-clan pairs; only cross-clan pairs survive: $(E_L v^{(2)})_i = \sum_{j \in A} X_{ij}$ for $i \in B$, and $(E_L v^{(2)})_i = \sum_{j \in B} X_{ij}$ for $i \in A$.*

Proposition 4 (Per-node variance). *Under Definition 4 and the observation model (15), with $\beta_0 = S_{\text{out}}^{\max}$ and using $(c_A + c_B)^2 = m/(ab) = (1 + \eta)^2/(m\eta)$,*

$$\sigma_B^2 := \text{Var}((E_L v^{(2)})_i)_{i \in B} = \frac{(1-p)(1+\eta)\beta_0^2}{\eta p}, \quad \sigma_A^2 = \frac{(1-p)(1+\eta)\beta_0^2}{p} = \eta \sigma_B^2, \quad (32)$$

so clan B has strictly smaller per-node variance; $\sigma_B^2 = (1-p)\tilde{\rho}(\eta)/p$ with $\tilde{\rho}(\eta) = (1+\eta)\beta_0^2/\eta$.

Proof. For $i \in B$ each of the a surviving terms has $S_{ij} = \beta_0$ and $|v_i^{(2)} - v_j^{(2)}| = c_A + c_B$, so $\sigma_B^2 = \frac{1-p}{p}(c_A + c_B)^2 a \beta_0^2$; substituting $a = m/(1+\eta)$ and $(c_A + c_B)^2 = (1+\eta)^2/(m\eta)$ gives the stated value. Clan A follows symmetrically with b in place of a . $\square$

Proof of Lemma 6. By Lemma 9 and proposition 3, for $i \in B$ the quantity $(E_L v^{(2)})_i = \sum_{j \in A} X_{ij}$ is a sum of independent, mean-zero summands with $|X_{ij}| \leq \beta_0(c_A + c_B)/p = R_B$ and total variance σ_B^2 (Proposition 4). Bernstein's inequality for bounded independent summands gives the stated two-term deviation bound at confidence $1 - \delta$. In the variance-dominated regime $p = \Theta(\log m/m)$ the linear term is of lower order, and a union bound over the m nodes at $\delta = \delta_0/m$ yields (26). $\square$

Proposition 5 (Clan B is binding). *Under Lemma 2, $\min_i |v_i^{(2)}| = |c_B|$, attained on the larger clan B. The per-node condition $|(E_L v^{(2)})_i| < \Delta \lambda |v_i^{(2)}|$ requires $p > 8m(1+\eta)\beta_0^2 \log m / \Delta \lambda^2$ on B (the clan-A condition carries an extra $1/\eta$), so since $\eta \geq 1$, clan B binds.*

Proof. Squaring $\sqrt{8\sigma_B^2 \log m} < \Delta \lambda c_B$ and substituting σ_B^2 with $c_B^2 = 1/(m\eta)$: the η cancels, giving $p > 8m(1+\eta)\beta_0^2 \log m / \Delta \lambda^2$. Clan A follows with $c_A^2 = \eta/m$. $\square$

Corollary 4 (Per-node recovery condition). *If $|w_i| < \min_i |v_i^{(2)}|$ for all i then $\text{sign}(\hat{v}^{(2)}) = \text{sign}(v^{(2)})$. Combining Lemma 6 and proposition 5 with the gap bound $\Delta \lambda \geq m(\rho - \eta\beta_0)/(1+\eta)$ (Lemma 3) yields*

$$p > \frac{8(1+\eta)^3 \beta_0^2 \log m}{m(\rho - \eta S_{\text{out}}^{\max})^2} = \Theta\left(\frac{\log m}{m}\right), \quad (33)$$

which is the rate (18) of Theorem 2. This computation is Step 4 of the proof in Section 6.

C.3 The second-order remainder

The bound below is the term discarded at Step 2 of Section 6; it is what restricts the argument to the strong-signal regime $\|E_L\|_2 \ll \Delta \lambda$.

Lemma 10 (Second-order negligibility). $\|w^{(2)}\|_\infty = O(\|E_L\|_2^2 / \Delta \lambda^2)$.

Proof. By Davis–Kahan $\|w\|_2 \leq \|E_L\|_2 / (\Delta \lambda - \|E_L\|_2)$. Each coefficient $b_\ell = -\langle v^{(\ell)}, E_L w \rangle$ satisfies $|b_\ell| \leq \|E_L\|_2 \|w\|_2$, and since $|\lambda_\ell - \hat{\lambda}_2| \geq \Delta \lambda - \|E_L\|_2$ each summand of $w^{(2)}$ contributes $O(\|E_L\|_2^2 / \Delta \lambda^2)$ in ℓ_∞ ; the $v^{(2)}$ component obeys $|\langle v^{(2)}, w \rangle| = O(\|w\|_2^2) = O(\|E_L\|_2^2 / \Delta \lambda^2)$. $\square$

D The Neumann trick — comparison with standard bounds

The recovery criterion (17) constrains $w = \hat{v}^{(2)} - v^{(2)}$ in the entry-wise sense, $\|w\|_\infty < c_B$, where $c_B = \min_i |v_i^{(2)}|$ is the entry-wise margin, attained on the binding clan B with $c_B^2 = 1/(m\eta)$, while the sampled-Laplacian noise satisfies $\|E_L\|_2 \leq \tilde{C} \sqrt{m \log m/p}$ (Lemma 4). The two routes below bound the same w , differing only in how they relate the entry-wise constraint to E_L .

The standard (ℓ_2) route. Davis–Kahan controls only the Euclidean norm, $\|w\|_2 \leq \|E_L\|_2 / (\Delta\lambda - \|E_L\|_2)$. Because the criterion is in the supremum norm, this must be transferred via $\|w\|_\infty \leq \|w\|_2$; imposing $\|w\|_2 < c_B$ and substituting the noise bound in the regime $\|E_L\|_2 \ll \Delta\lambda$ yields $p_{\text{naive}} \gtrsim m^2\eta \log m / \Delta\lambda^2$, i.e. $\eta(1+\eta)^2 \log m / (\rho - \eta\beta_0)^2$ after the gap bound of Lemma 3.

The entry-wise (Neumann) route. Lemma 5 bounds each coordinate directly, replacing $\|E_L\|_2$ by the scalar $(E_L v^{(2)})_i$ at the tightest coordinate $i \in B$. This scalar is a sum of $O(m)$ independent mean-zero contributions concentrating at $\sigma_B^2 = (1-p)(1+\eta)\beta_0^2/(\eta p)$ (Proposition 4), smaller than $\|E_L\|_2^2$ by a factor m ; imposing $|w_i| < c_B$ and the gap bound recovers the rate of Corollary 4.

Remark 2 (ℓ_2 vs. entry-wise). Comparing the two rates gives $p_{\text{naive}}/p_{\text{Neumann}} = \Theta(m\eta/((1+\eta)\beta_0^2))$, a $\Theta(m)$ reduction for fixed η . The saving originates at the noise level: the operator norm $\|E_L\|_2$ exceeds the individual increment $(E_L v^{(2)})_i$ by $\Theta(\sqrt{m})$, and since p enters both rates as p^{-1} under a square root this squares to $\Theta(m)$. Both rates carry $(\rho - \eta\beta_0)^{-2}$ and so diverge together as $\rho - \eta\beta_0 \rightarrow 0$ (the boundary of Condition C1), but the entry-wise rate stays within $p \leq 1$ over a region where p_{naive} already exceeds it, so it accommodates more imbalanced trees than the Euclidean route admits.

E Which Split Each Operator Picks

Two cohorts of 1000 trees, Kingman and birth–death, at $m = 1000$ and $L = 10^4$ under Assumptions A1 and A2, with no sub-sampling. Each operator uses the rounding rule its own theorem states: k -means on the Fiedler vector of $L(S)$, sign on the leading-magnitude eigenvector of $\mathcal{B} = HDH$. Per tree we record the imbalance of the returned split, $\eta = \max(n_1, n_2) / \min(n_1, n_2)$, and whether that split is an edge of the true tree. Figure 7 reports both, per cohort and per operator.

Table 1: Both operators on both cohorts, 1000 trees each.

	Kingman		birth–death	
	$L(S)$	$\mathcal{B}$	$L(S)$	$\mathcal{B}$
valid single-edge split	996	986	996	555
median η	3.00	1.85	3.08	1.53
maximum η	499	24.6	999	5.3
only $L(S)$ valid		14		443
only $\mathcal{B}$ valid		4		2
identical η		64.1%		33.2%

Table 1 summarises the two cohorts. Four things follow.

1. **On Kingman trees the operators differ in *which* edge, not in whether they find one** — 99.6% and 98.6% valid, and the same split on 64.1% of trees.
2. **$\mathcal{B}$ ’s output is confined to near-balanced splits:** $\eta \leq 24.6$ on Kingman and $\eta \leq 5.3$ on birth–death, against true splits reaching $\eta = 999$. This is ?? — each edge contributes at least $2\tau_e n_1(e)n_2(e)/m$ to $|\lambda_1(\mathcal{B})|$, and $n_1(e)n_2(e)$ is maximal at balance. Where the rules disagree, $\mathcal{B}$ ’s split is the more balanced one in 355 of 359 Kingman cases and 654 of 668 birth–death cases.

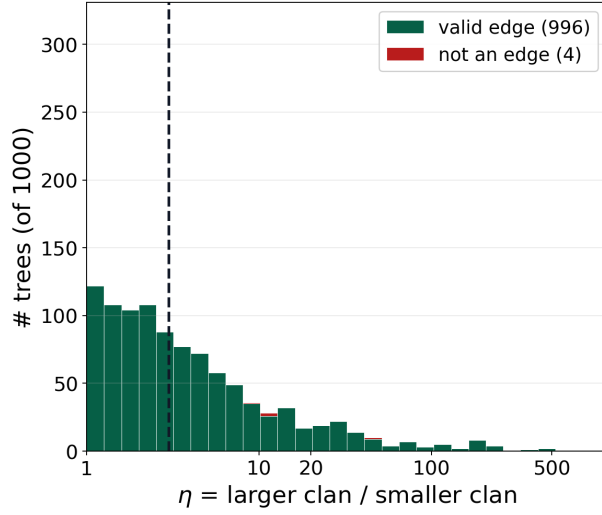
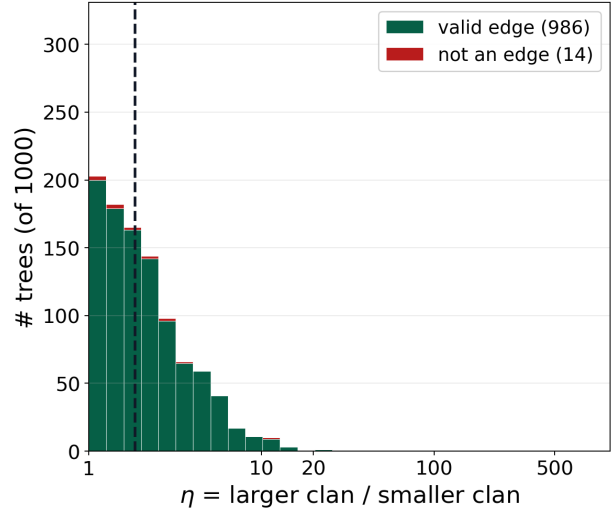
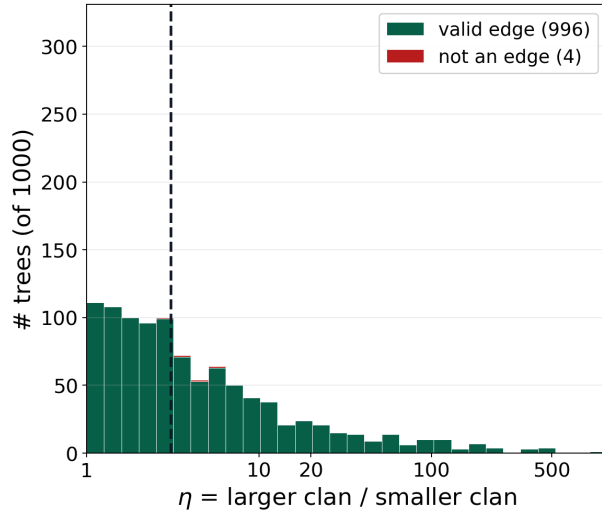
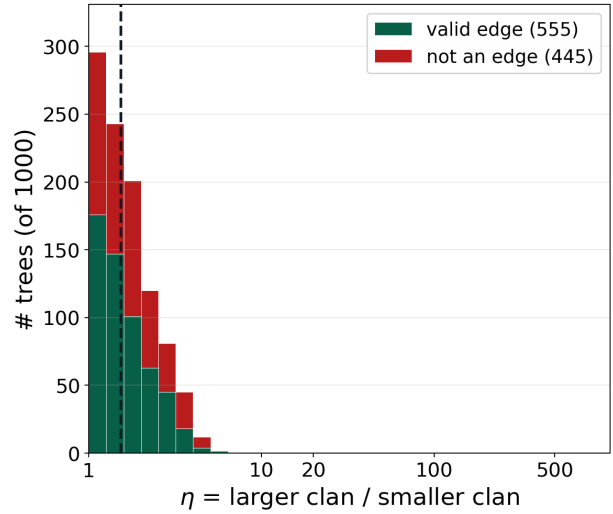
(a) Kingman, $L(S)$, k -means(b) Kingman, $\mathcal{B}$, sign(c) birth-death, $L(S)$, k -means(d) birth-death, $\mathcal{B}$, sign

Figure 7: **Imbalance of the returned split, 1000 trees per model.** Green is a genuine single-edge bipartition of the true tree, red is not; dashed line is the median. All four panels share bins and axes.

3. **That confinement breaks $\mathcal{B}$ on imbalanced trees.** Its validity falls from 98.6% to 55.5% while $L(S)$ holds at 99.6%; on 443 birth–death trees $L(S)$ returns an edge and $\mathcal{B}$ does not. Lacking a near-balanced edge it does not decline — it returns a near-balanced *non*-edge.
4. **A narrow η distribution is a property of the operator, not of the tree.** Comparing the two routes’ sampling rates is meaningful only with the target split held fixed; otherwise the rates describe different recovery problems, and on birth–death trees one is conditioned on a split absent from the tree.